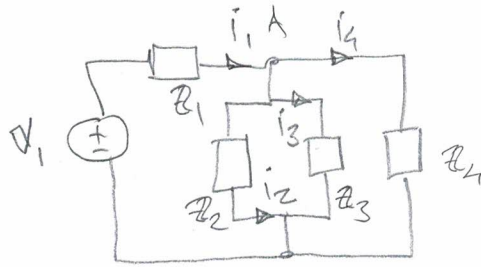


4.50)



$$\begin{aligned} i_1 &= 150 \cos(\omega t + 4) \\ i_2 &=? \\ i_3 &= 80 \sin(\omega t - 1.2) \\ i_4 &= 150 \sin(\omega t + 2) \end{aligned}$$

Beside  $i_2$ .

Wdh: for node A give KCL  $i_1 - i_4 - i_2 - i_3 = 0$

$$\Rightarrow \underline{i_2 = i_1 - i_3 - i_4}$$

SKRIP  $\cos(\omega t + 4) = \sin(\frac{\pi}{2} - \omega t - 4) = -\sin(\omega t + 4 - \frac{\pi}{2})$

SKRIP IN KOMPLEX FORM

$$I_2 = I_1 - I_3 - I_4 = -150e^{j(\omega t + 4 - \frac{\pi}{2})} - 80e^{j(\omega t - 1.2)} -$$

$$-150e^{j(\omega t + 2)} =$$

$$= e^{j\omega t} \left( -150e^{j(4 - \frac{\pi}{2})} - 80e^{-j1.2} - 150e^{j2} \right) =$$

$$= e^{j\omega t} \left( -150 \cos(4 - \frac{\pi}{2}) - 150j \sin(4 - \frac{\pi}{2}) - \right.$$

$$\left. -80 \cos(1.2) - j80 \sin(1.2) - 150 \cos(2) - \right.$$

$$\left. -j150 \sin(2) \right) = e^{j\omega t} (109.11 - j127.196)$$

$$= e^{j\omega t} 167.58 e^{-j49.38^\circ} = e^{j\omega t} 167.58 e^{-j0.862}$$

$$= 167.58 e^{j(\omega t - 0.862)}$$

TAJ MAGNITUDE DAN  $\angle$   $I_2$  SIA FAS

$$i_2(t) = 167.58 \sin(\omega t - 0.862) \text{ A}$$

SVAR:  $i_2(t) = 167.58 \sin(\omega t - 0.86) \text{ A}$